



ZION ENERGY SERVICES

DS-1 INSPECTION MEETING / TRAINING

Audit follow-up | Required inspection steps

Meeting ID: _____ Date: _____ Time: _____

Location / crew: _____

Trainer: _____ Customer / job: _____

Controlled DS-1 Vol. 4 edition / addenda: _____

Purpose: Review the required inspection steps identified in the audit and document the meeting, training provided, and employee attendance.

01 BOX SWELL | 7.14.5.6

Place a straightedge lengthwise along the box OD. If there is a visible gap, use calipers to compare the OD at the bevel with the OD 2 inches (+/- 1/2 inch) away. Reject the connection when the OD at the bevel is larger by 1/32 inch or more.

Reviewed: _____ Demonstrated: _____ Trainer initials: _____

02 SHOULDER FLATNESS | 7.14.5.15

On the box, place a straightedge across a diameter of the face and rotate it at least 180 degrees in the shoulder plane. Repeat on the pin with the straightedge across a chord of the shoulder. Any visible gap between the straightedge and shoulder is cause for rejection.

Reviewed: _____ Demonstrated: _____ Trainer initials: _____

03 BOX OUTSIDE DIAMETER | 7.16.4(a)

Measure box OD 4 inches (+/- 1/4 inch) from the shoulder. Take at least two measurements spaced 90 degrees (+/- 10 degrees) apart.

Apply the appropriate acceptance criteria: manufacturer OD requirements; Table 7.38 for components connected to HWDP; or customer-specified bending strength ratio (BSR), using box OD with pin ID, for drill collars or similar components. Refer to Tables 7.37 and 7.50 as applicable. Manufacturer criteria govern conflicts.

Reviewed: _____ Demonstrated: _____ Trainer initials: _____

Trainer demonstration: Show the two different measurement locations used for box swell and box OD, and demonstrate straightedge placement on both shoulders.



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Training topics continued | Meeting ID: _____ Date: _____

04 PIN STRESS RELIEF GROOVE (SRG) | 7.16.4(e)

Review applicability: Unless waived by the customer, API connections NC38 and larger that make up to BHA components, including applicable saver-sub connections, require a pin SRG; reject when the required feature is absent. Review Table 7.3 for product-specific stress relief requirements and its application notes.

Measure groove diameter and width against Table 7.37 or 7.38, as applicable. Measure groove length from the connection shoulder to the first full thread, with a metal rule on the thread taper and squared against the shoulder.

Identify the first full thread: It is nearest the pin shoulder and has the same height and profile as the second thread. Rotate the profile gauge until the least possible light appears between the thread form and gauge. Review Figures 7.51, 7.52, and 7.53 during the demonstration.

Reviewed: _____ Demonstrated: _____ Trainer initials: _____

05 MAGNETIC FIELD MAGNITUDE / ORIENTATION | 7.14.4(e)

Verify the field under blacklight with a magnetic flux indicator strip or magnetic penetrameter on the internal connection surface while solution is being applied and power is activated.

Perform the verification:

1. At the start of each inspection.
2. After every 25 connections.
3. Whenever component diameter, thickness, or length changes.
4. When the inspector changes.
5. Whenever the inspector or customer representative requests confirmation of proper magnetization.
6. Upon completion of the inspection.

Reviewed: _____ Demonstrated: _____ Trainer initials: _____

06 DISCUSSION / FOLLOW-UP ACTIONS

Questions, additional coaching, or actions needed:

Action owner: _____ Due date: _____

Training summaries are based on the supplied DS-1 pages. Review the controlled standard, relevant tables, customer requirements, and approved inspection procedure during the meeting.



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Employee acknowledgment | Keep this page with the training topics

Meeting ID: _____ Date: _____ Location: _____

Trainer / meeting leader: _____

07 EMPLOYEE ATTENDANCE & TRAINING ACKNOWLEDGMENT

By signing below, I confirm that I attended this meeting and received training on the topics marked as reviewed on pages 1-2. I had an opportunity to ask questions and seek clarification on the required inspection steps.

This signature documents attendance and training received. Any individual competency evaluation or qualification is recorded separately.

No.	Printed employee name	Employee signature	Date
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			

08 MEETING LEADER CONFIRMATION

I confirm that this meeting was held on the date recorded above and that the topics marked as reviewed were covered with the employees who signed this record.

Leader printed name: _____

Leader signature: _____ Date: _____

Attachments / demonstration records: _____